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PROCESS CONCEPTS

2.1 Program Versus Process

Program	Process
Program is a set of instructions and data.	A program under execution is called a process.
It resides in the secondary memory.	It resides in the main memory.
It is passive.	It is active and dynamic.
It is not allocated any resources.	The operating system allocates resources to the process for its execution.

2.2 Process as ADT

The process image can be described as:

Process		
Process attributes (Process Control Block)	}	Context
Run time Stack		
Dynamic data	}	User address space
Static data		
Code section		

Any process has the following attributes:

- Process identification information
- Priority
- Process state information
- Program counter
- Memory limits
- List of files
- List of open devices
- Protection information

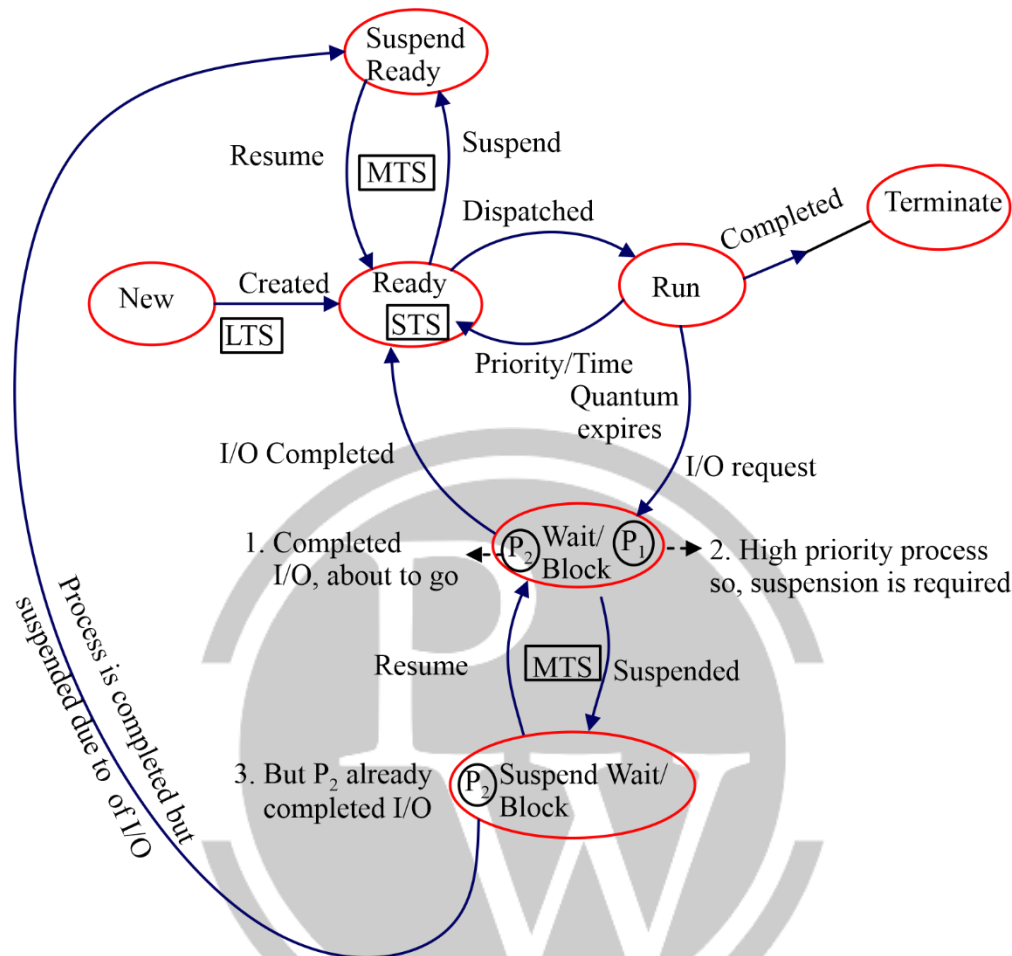
All the attributes of the process are stored in the Process Control Block (PCB). The features of Process Control Block are as follows:

- Every process has its own process control block.
- The process control blocks of all the processes are stored in the main memory.
- They are implemented using **double linked list** data structure.

The **context** of a process includes the process attributes and the stack information.

2.3 Process State Transition Diagram

A process passes through multiple states in its lifetime as follows:



- When the process is present in the Ready, Run or Wait state, it still resides in the main memory.
- When the process is present in the Suspend Ready or Suspend Block state, it resides in the secondary memory.
- There can be multiple processes present simultaneously in the Ready, Wait, Suspend Ready and Suspend Wait states.
- In the Run state, only one process can exist at a time.

2.4 Schedulers

The operating system deploys three kinds of schedulers:

- Long-term scheduler.
- Medium-term scheduler.
- Short-term scheduler.

2.4.1 Long-term scheduler

- It is responsible for the creation and bringing of new processes into the main memory.
- It controls the degree of multi-programming.
- It should generate a good mix of CPU and I/O bound process for efficient resource utilization.
- It is involved in the New → Ready state transition.

2.4.2 Medium-term scheduler

- The medium-term scheduler acts as the swapper by doing swap-out (suspending the process from the main to secondary memory) and swap-in (resuming the process by bringing it from the secondary to main memory) operations.
- It is involved in Ready $\Rightarrow$ Suspend Ready and Block $\Rightarrow$ Suspend Block state transitions.

2.4.3 Short-term scheduler

- The short-term scheduler selects a process from the ready state to be executed.
- It is involved in the Ready $\rightarrow$ Run state transition.

2.5 Dispatcher

- The dispatcher is responsible for loading the job (selected by the short-term scheduler) onto the CPU. It performs context switching. Context switching refers to saving the context of the process which was being executed by the CPU and loading the context of the new process that is being scheduled to be executed by the CPU.

